

Appendix

1 Prompts

Classify Question (I_C)

Classify a given question as either "self-contained" or "non-self-contained" while considering the context of resolving pronouns and references. In this task, "self-contained" questions are those that can be understood and answered without needing additional context or information, even when they refer to entities or concepts mentioned in prior conversation. "Non-self-contained" questions are those that, due to the presence of pronouns or references to prior conversation, require additional context or information to be answered correctly.

Question: {question}
Classification:

Rephrase Ambiguous Question(I_{RF})

Given the following conversation and a follow up question, rephrase the follow up question to be a standalone question. Preserve the original question in the answer sentiment during rephrasing.

Chat History: {chat_history}
Follow Up Input: {question}
Standalone question:

Question Understanding (I_{QU})

Extract triples from the given question by following these steps carefully.
****Let's think step by step.****

Step 1: Extract Key Entities

- Identify explicit entities mentioned in the question.
- If an entity is missing but implied, introduce a variable (e.g., ?var).

Step 2: Identify the Predicates

- ****Prioritize nouns that represent the core relationship being questioned.**** These nouns often describe the key property or attribute being asked about (e.g., "capital," "author," "moons").
- ****Explicitly exclude generic verbs like "has," "is," "are," "was," "were," "do," "does," "did" when a more specific noun or verb can express the relationship.****
- If no such core noun exists, then consider other verbs, excluding the helper verbs listed above.
- Extract the most relevant predicate from the question.
- The predicate should be specific and relates two entities.

Step 3: Construct the Triples

- Formulate the triple using the extracted subject, predicate, and object.
- If the object is unknown, use a variable (e.g., ?answer).
- Ensure at least one meaningful triple is produced.

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### **Step 4: Output the Result**
- Return the final output as a JSON object in this exact format: ‘{ "triples": [] }’.

**Let’s apply this step-by-step approach to extract the correct triple.**

### **Examples:**
#### **Example 1**
**Question:** Who wrote Pride and Prejudice?
**Step 1 - Extract Entities:** ["Pride and Prejudice", "?author"]
**Step 2 - Identify Predicates:** "wrote"
**Step 3 - Construct Triples:** ‘[ ["?author", "write", "Pride and Prejudice"] ]’

#### **Example 2**
**Question:** What is the capital of France?
**Step 1 - Extract Entities:** ["France", "?capital"]
**Step 2 - Identify Predicates:** "capital"
**Step 3 - Construct Triples:** ‘[ ["France", "capital", "?capital"] ]’

### **Input:**
Question: {question}

### **Response:**
‘‘‘json

```

Vertex Selection (I_{VL})

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#### Instruction:
Given an entity and a list of labels, return the most similar label.
1. If an exact match (case-insensitive) exists in the list, return that label immediately and
exclusively, ignoring any other labels.
2. If no exact match is found, select the most semantically similar label based on meaning,
context, and common associations.

**Important Notes:**
- An **exact match** means the entity must match a label character-for-character, ignoring case.
- Do not modify, append, 3or alter the exact-match label.
- If an exact match exists, it must be returned even if a semantically similar label is also
present.
- Prioritize labels that closely match the entity meaning, not just syntactic similarity.

Only Return the label in a **JSON object** in the following format: {"value": label}.
Do not add any explanations.

### Inputs:
Entity: {entity}
List of labels: [{vertex_label_list}]

### Response:‘‘‘json

```

Query Selection (I_{QS})

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### Instruction:
Let’s think step by step about what the question is asking and what the answer would look like.
- Evaluate each keyword relevance to retrieving the correct answer. Focus on keywords that are
directly related to answering the question.

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- Only include keywords that directly contribute to retrieving the correct answer.
- Exclude keywords related to the item or context, and focus solely on those that provide the key information needed for the answer.

Only choose keywords from the provided list - do not generate new ones.

Return the keywords in a JSON object in the following format:

```
{{ "keywords": [] }}.
```

Inputs:

Question: {question}

List of Keywords: [{predicate_list}]

Response: ``json

Final Answer Reformalization I_{FA}

Given a question and its structured answer extracted from the Knowledge Graph. Convert the structured answers to an equivalent natural language answer suitable for human conversation

Question: {question}

Answers: {answers}

Reformalized Answer:

Question Translation I_{Tr}

If the following question is in English, output it exactly as it is. If it is in another language, translate it into English.

Question: {question}

Output:

2 Error Analysis

Improved question understanding directly impacts downstream accuracy, but gains are not always linear due to failures in other pipeline stages. The remaining errors include structural mismatches, entity linking, and query selection. In QALD-9, many errors result from structural mismatches between natural language expressions and the organization of the underlying KG. For instance, questions like “*How many emperors did China have?*” require linking to entities such as *Emperor of China*, rather than to *China* itself. Similarly, “*Who killed Caesar?*” should retrieve *Assassins of Julius Caesar*, not just *Caesar*. Further challenges arise when questions involve vague or generic KG predicates, such as *subject*, which lack semantic clarity. These structural mismatches represent cases where the question understanding is correct, but the KG’s organization differs from how the information is naturally expressed in language.

Entity linking failures occur when the system selects an incorrect vertex from multiple candidates, often due to ambiguity in entity names or semantic similarity between KG vertices. For example, in the question “*Are Taiko some kind of Japanese musical instrument?*”, the system links to http://dbpedia.org/resource/Japanese_musical_instrument, while the correct entity in the graph is <http://dbpedia.org/class/yago/WikicatJapaneseMusicalInstruments>. Query selection failures are categorized into two reasons: 1) the agent decides that all candidate queries are irrelevant to the question and returns nothing or 2) it selects a query with an incorrect predicate. For instance, in “*How many grandchildren did Jacques Cousteau have?*”, the agent incorrectly selects the *relative* predicate instead of *child*. These failures often cascade: incorrect entity linking produces weak candidate queries, which the query selection agent may then reject or misrank. While Chatty-KG includes a retry mechanism to mitigate some of these failures, retries are limited to three attempts to control latency and cost. Despite these challenges, Chatty-KG significantly reduces both overall and QU-specific failures.

3 Multilingual Support

In this experiment, we extended the CHATTY-GEN pipeline to handle multilingual questions by implementing an LLM-based translation module that translates non-English questions to English before processing begins. This design enables a single unified pipeline for all languages, as the translation module passes English questions through unchanged, demonstrating the flexibility and extensibility of CHATTY-GEN. We chose to translate questions before processing because the underlying hosted knowledge graph is in English and the SPARQL queries use English entities. To evaluate CHATTY-GEN’s multilingual capabilities, we used the QALD-9 dataset, which provides each question in eleven different languages in addition to English, as shown in Table 1. The results demonstrate that the system maintains consistent performance across languages, with F1 scores ranging from 42.08 for Persian to 50.79 for German compared to 49.33 for English. Interestingly, some languages such as German and Portuguese slightly outperformed English. Analysis revealed that translation often normalized and clarified entity references, for instance simplifying "The Scream by Munch" to "Munch’s The Scream" and disambiguating "Free University of Amsterdam" to "Vrije Universiteit Amsterdam," which facilitated more accurate entity linking. Languages with lower performance, such as Persian, can be attributed to variations in translation quality across different source languages.

Table 1: Results of running CHATTY-GEN with the translation extension module on the 11 languages of QALD-9

Language	P	R	F1
English(en)	62.29	40.83	49.33
German(de)	66.31	42.16	50.79
Portuguese(pt)	63.62	42.14	50.70
Hindi(hi)	65.20	39.36	49.08
French(fr)	64.21	37.72	47.52
Dutch(nl)	65.80	36.40	46.87
Italian(it)	62.97	37.14	46.72
Romanian(ro)	65.64	36.05	46.54
Spanish(es)	64.37	36.16	46.31
Russian(ru)	67.95	32.71	44.16
Persian(fa)	68.69	30.33	42.08

3.1 Examples

Questions from QALD-9 dataset clarified and solved after translation to English

3.1.1 German

- **Question ID: 192**

- **English:** Which museum exhibits The Scream by Munch?
- **German:** In welchem Museum ist Der Schrei von Munch ausgestellt?
- **Translated:** In which museum is Munch’s The Scream exhibited?
- **Comment:** The correct Entity is *The Scream*, in the English version the entity may be confused to be *The Scream by Munch*, however, the version translated from German clarified it.

- **Question ID: 212**

- **English:** Which companies work in the aerospace industry as well as in medicine?
- **German:** Welche Unternehmen sind sowohl in der Luft- und Raumfahrt als auch im Bereich der Medizin tätig?
- **Translated:** Which companies are active in both aerospace and the medical field?
- **Comment:** The more straightforward conjunction *and* made it easier for the system to understand the question.

3.1.2 Portuguese

- **Question ID: 10**

- **English:** How many students does the Free University of Amsterdam have?
- **Portuguese:** Quantos estudantes a Universidade Livre de Amsterdã tem?
- **Translated:** How many students does the Vrije Universiteit Amsterdam have?
- **Comment:** The English question requires semantic understanding that the *Free University of Amsterdam* is the *Vrije Universiteit Amsterdam*. However the version translated from portuguese clarifies it.

4 Human and LLM Evaluation

To assess the conversational quality of CHATTY-KG, we conducted a comprehensive comparative evaluation involving **five evaluators**: three human annotators and two large language models (CHATGPT and GEMINI). The process followed a structured two-part protocol. In *Part A*, each question–answer pair within a dialogue was independently rated on two dimensions—*Response Quality* and *Fluency*. Response Quality captured informativeness, relevance, and consistency with prior context, while Fluency reflected grammaticality, naturalness, and stylistic appropriateness. In *Part B*, after reviewing an entire dialogue, evaluators provided three additional scores at the conversation level: *Referential Coherence*, *Topic Continuity*, and *Logical Flow*. All ratings used a 1–5 integer scale, where 1 indicated poor performance and 5 represented excellence. For analysis, we computed the average score per dimension by aggregating the corresponding column groups (e.g., all content-level metrics for Quality), resulting in the summary statistics shown in Table 2.

Detailed results for each individual evaluation factor before aggregation are presented in Table 3, which highlights fine-grained differences across human and LLM assessments. The complete evaluation instructions followed by both human annotators and language models are shown in the box below, ensuring transparency and reproducibility of the scoring process.

As shown in Table 2, both human and LLM evaluators exhibited strong alignment in their assessments. The system achieved an average of **4.45** in Response Quality, **4.86** in Fluency, and **4.51** in Dialogue Coherence, indicating consistently high performance across all dimensions. The standard deviations remained low (0.15, 0.06, and 0.04, respectively), confirming stable and repeatable judgments. To further quantify inter-rater reliability on ordinal ratings, we computed **Weighted Cohen’s Kappa** (κ), yielding $\kappa = 0.53$ for human–human and $\kappa = 0.56$ for human–LLM agreement. These values indicate *moderate agreement*, which is appropriate given the skewed distribution of high scores (majority rated as 5), and is supported by the consistently high means and low variance.

Overall, these results confirm that both human and LLM evaluators perceive CHATTY-KG’s responses as highly fluent, coherent, and contextually appropriate. The agreement metrics demonstrate that conversational quality is reliably assessed across evaluators, supporting the system’s ability to balance linguistic naturalness with content accuracy and dialogue flow.

Human and LLM Evaluation Instructions

Chatty-KG Evaluation Instructions

- Each dialogue (marked by 'Did') represents a short conversation about one entity, such as a person or place.
 - Each row inside that dialogue is a single question and its final answer.
- You will rate every question individually, and then give three overall scores for the whole dialogue.

Part A - Per-Question Evaluation

(Columns E-J - one score per cell for every question)

For every question-answer pair, you should score two categories: Response Quality and Fluency.

Use integer values from 1 (very poor) to 5 (excellent).

Response Quality

Judge how well the answer content fits the question. Sample scores are given below.

Informativeness: Does the answer contain meaningful information or is it vague?

- Score 1 if it says things like "not available," "no structured answer," or gives empty output.
- Score 3 if it gives partial but not full information.

- Score 5 if it is clear and complete.

Relevance: Is the answer directly addressing what the question asks?

- Score 1 if it drifts or misunderstands the question.
- Score 3 if it partly answers it.
- Score 5 if it is fully on target.

Consistency with Context: If the dialogue has earlier turns, does this answer agree with them?

- Score 1 if it contradicts earlier answers.
- Score 3 if neutral or uncertain.
- Score 5 if it remains fully consistent.

Fluency

Judge how natural and grammatical the text sounds. Sample scores are given below.

Grammaticality: Rate the correctness of grammar and structure.

- 1 means ungrammatical or broken English.
- 3 means minor issues but understandable.
- 5 means fully correct.

Naturalness: Does the answer read smoothly, like human speech rather than a template?

- 1 means robotic or awkward.
- 3 means acceptable but mechanical.
- 5 means fluent and natural.

Style Fit: Is the wording concise and appropriate for a factual Q&A?

- 1 means too verbose, casual, or poorly formatted.
- 3 means acceptable style.
- 5 means clean, direct, and professional.

You will therefore fill six numeric values (one for each of the above factors) for every question-answer pair.

Part B - Per-Dialogue Evaluation

(Columns K-M - filled once for the whole dialogue block)

After rating all individual questions in one dialogue, read the entire set together and provide three overall scores that describe how coherent the conversation is.

Use the same 1-5 scale.

Referential Coherence: Across turns, are pronouns like "he," "she," or "it" correctly resolved so that the reader always knows who or what is being discussed?

- Give 1 if references are confusing or wrong.
- Give 3 if they are sometimes clear.
- Give 5 if they are always correct and easy to follow.

Topic Continuity: Does the dialogue stay on the same entity or drift to another topic?

- Give 1 if it switches topics or entities.
- Give 3 if it has minor topic shifts.
- Give 5 if it remains consistent from start to end.

Logical Flow: Do the answers form a logical progression without contradictions or abrupt resets?

- Give 1 if the sequence is inconsistent.
- Give 3 if it partly follows logically.
- Give 5 if it flows smoothly across the turns.

Enter these three coherence scores only once per dialogue, ideally in the first row of that block.

Scoring Guidelines

- Use only whole numbers from 1 to 5.
- If you are unsure, choose 3 as a neutral score.

Table 2: Human and model evaluation across quality dimensions, including inter-annotator agreement.

Metric	ChatGPT	Gemini	Human 1	Human 2	Human 3	Avg	Std
Response Quality	4.48	4.64	4.50	4.43	4.22	4.45	0.15
Fluency	4.92	4.86	4.88	4.76	4.86	4.86	0.06
Dialogue Coherence	4.52	4.52	4.50	4.45	4.57	4.51	0.04
Weighted Cohen’s κ (Human–Human):			0.53 (Moderate Agreement)				
Weighted Cohen’s κ (Human–LLM):			0.56 (Moderate Agreement)				

Table 3: Detailed Human and model evaluation across content-, language-, and conversation-level dimensions.

Model / Human	Response Quality (content-level)			Fluency (language-level)			Dialogue Coherence (conversation-level)		
	Informativeness	Relevance	Consistency with Context	Grammaticality	Naturalness	Style Fit	Referential Coherence	Topic Continuity	Logical Flow
ChatGPT	4.00	4.59	4.86	4.99	4.89	4.88	4.65	4.75	4.15
Gemini	4.15	4.83	4.94	4.98	4.80	4.81	4.45	4.90	4.20
Human 1	4.03	4.61	4.86	4.99	4.79	4.87	4.65	4.75	4.10
Human 2	4.00	4.59	4.71	4.86	4.65	4.77	4.65	4.58	4.12
Human 3	3.77	4.40	4.48	4.89	4.81	4.87	4.80	4.55	4.35

- You may add a short comment at the end of a dialogue if something unusual occurs (for example, "vague answers" or "reference unclear").
- After finishing all rows, compute averages per dialogue if needed.

5 Structured Query Generation vs. RAG

Retrieval-Augmented Generation (RAG) systems aim to enhance large language models (LLMs) by grounding their responses in external knowledge retrieved from textual or semi-structured corpora. Instead of relying solely on parametric memory, a RAG pipeline retrieves relevant passages from an indexed collection, feeds them to the LLM, and lets it synthesize an answer in natural language. While this approach improves factual accuracy in many open-domain tasks, it often struggles when precise, structured answers are required. GraphRAG represents an advanced variant of RAG that retrieves from graph-based data sources, attempting to combine retrieval over graph structures with LLM-based reasoning to generate knowledge-grounded responses.

To evaluate this capability, we applied GraphRAG to the DBLP benchmark over the DBLP knowledge graph, focusing on structured question answering. The system achieved an average **precision of 46.00**, **recall of 37.01**, and **F1-score of 38.99**, reflecting moderate precision but limited completeness. The analysis revealed several consistent patterns: (1) when the gold answer was a long list of entities, the system often failed to retrieve any results, returned only a small sample of the expected set (thus reducing recall), or generated a narrative summary that hallucinated or paraphrased the true content rather than listing structured answers; (2) for questions requiring multiple reasoning hops, GraphRAG, in most cases, failed to connect the relevant entities, likely because the text chunking process disrupted the semantic connectivity between related nodes; (3) for simple single-triple questions, the system achieved perfect scores when the target fact was explicitly represented in the knowledge base; and (4) when entity labels were missing and only IRIs were available, the model tended to produce free-text, LLM-based answers instead of structured triples.

In summary, GraphRAG demonstrates strong precision for directly retrievable facts but faces challenges in scaling to multi-hop reasoning and large answer sets, where retrieval fragmentation and narrative generation degrade both grounding and recall. These results highlight the trade-off between the flexibility of language models and the structural rigor required for accurate, query-based reasoning over knowledge graphs.

6 Discussion

Flexibility: CHATTY-GEN is a modular, domain-agnostic architecture that can work on KGs across different domains. Its KG-agnostic matching agent does not require any prior knowledge of the KG structure or schema, instead, it dynamically extracts relevant subgraphs by executing SPARQL queries. This design enables CHATTY-GEN to operate on specialized domain KGs without requiring pre-processing, training, or domain-specific adaptation.

Lightweight Training vs Prompting: The different agents of Chatty-KG operate using prompting strategies tailored to task complexity to guide LLM outputs—zero-shot for simpler tasks such as the Rephraser agent, few-shot prompting for more complex agents such as the QIR agent, and chain-of-thought reasoning where appropriate. This training-free

Table 4: Number of questions solved by CHatty-KG across different question types in KGQA benchmarks

Benchmark	Factoid	Count	Boolean	Q Len	Preds
QALD-9	63/133	2/13	0/4	7.52	1.77
YAGO-B	77/99	-	0/1	6.97	2.00
DBLP-B	70/92	7/8	-	9.73	1.72
MAG-B	58/81	16/17	2/2	8.63	1.87
DBpedia-Dialog	34/64	0/11	6/19	7.10	1.00
YAGO-Dialog	48/65	0/3	10/18	8.21	1.02
DBLP-Dialog	48/59	2/4	17/27	11.62	1.01
Wikidata-Dialog	36/61	1/5	15/25	9.05	1.03

design enables rapid deployment without waiting for dataset generation or fine-tuning to complete. Although lightweight fine-tuning or retrieval tuning could potentially improve performance, these approaches require task- and KG-specific datasets for each subtask (e.g., question understanding, query generation), reducing the system’s flexibility to operate across arbitrary and evolving graphs without prior customization. Empirically, our prompting-based approach demonstrates competitive performance: Chatty-KG achieves better results than KGQA—which uses fine-tuning for question understanding—through prompting alone as shown in Figure 3. While fine-tuned models may offer task-specific gains, prompting provides superior scalability, generalizability across diverse KGs, faster deployment, and eliminates the need for specialized training datasets, making Chatty-KG more practical for real-world applications involving heterogeneous and evolving KGs.

Key Learnings: Building Chatty-KG as a multi-agent system for KGQA yielded several insights. (1) The modular agent architecture enabled iterative improvements of individual modules without requiring system-wide modifications, facilitating system development. (2) Agent coordination through a shared state allows agents to determine execution flow by consulting a common context. (3) Incorporating validation mechanisms at the agent level significantly reduced hallucinations by catching invalid outputs before propagation. (4) Any retry mechanisms used with LLMs must be bounded to prevent infinite loops when tasks exceed LLM capabilities. (5) The multi-agent approach introduced minimal latency overhead, unlike initial expectations of increased latency. However, debugging complexity increased due to distributed execution. These design choices prioritize accuracy and modularity while maintaining practical performance.

Question Type Analysis KG-based QA systems are best suited for fact-based questions such as who, what, where, when, count, list, and compare. They are less effective for open-ended questions like why or how, which often require reasoning beyond the scope of a KG. Table 4 presents the distribution of question types across the different KGQA benchmarks and the performance of Chatty-KG across these categories.

Our developed taxonomy categorizes questions into three types based on the linguistic nature of the question by mapping question words and patterns to the following categories:

1. **Factoid questions:** answers are entities or literal values existing in the KG. These are the most common question type in KGQA benchmarks. This category includes questions beginning with: *Who, Whom, What, Which, Where, When, and How*. It also encompasses imperative forms (*List, Give, Show, Return, Name, State*) and questions starting with noun phrases or named entities.
2. **Count questions:** require counting operations to get the answer. These are identified by the question phrase *How many*.
3. **Boolean questions:** expect yes/no answers and involve verification of facts in the KG. This category includes questions beginning with: *Is, Are, Was, Did, Does, Can, and Has*.

As shown in Table 4, Chatty-KG successfully handles the diverse question types present in KGQA benchmarks. For factoid questions, which constitute the majority of queries, the system achieves strong performance across all datasets. Count and Boolean questions show varied performance depending on their complexity and the benchmark characteristics.

The average question length in words (**Q Len**) and average number of predicates per SPARQL query (**Preds**) provide additional context about query complexity. Dialogue benchmarks show fewer predicates compared to basic benchmarks because conversational questions rely on dialogue history for context, reducing the need to explicitly specify all facts in each individual query.